BKEMS Prototype Store - Bench Wiring (draft rev 3, Oct 5 2026)

Rev 3: store node + ESP32 at EACH RTU. Buck fed from RTU transformer via ~1A inline fuse. C required (taken at the unit). Pins marked *(proposed)* are suggestions; relay pins and wire colors are from Andy's notes. MQTT contract v1.0.

POWER (store node)

5V bench supply (USB or breadboard PSU) feeds store ESP32 + Inland 4-ch relay

12V DC supply - *only if the 16-ch / 8-ch relay boards are used (not the 4-ch)*

5V

5V

12V (big boards only)

DS18B20 one-wire bus (GPIO4, *proposed*)

1. Walk-in cooler 2. Freezer
3. Dining room 4. Duct supply air

Red lead = 3V3

Black lead = GND

Yellow lead = data (GPIO4)

4.7k pull-up resistor: data to 3V3

Photocell / light sensor
analog out to GPIO34 (*proposed*)

AD2000-M door panel
status / alarm output
(panel keeps doing the locking)

Optocoupler
door signal
(*proposed*)

ESP32 DevKit on Freenove breakout
STORE NODE - device key store-{id}-gw

5V / VIN	power in from 5V supply
3V3	sensor power + pull-up
GND	sensor ground
GPIO4	DS18B20 data (*proposed*)
GPIO34	light analog (*proposed*)
GPIO32	door status (*proposed*)
relay IN1	**GPIO25**
relay IN2	**GPIO26**
relay IN3	**GPIO27**
relay IN4	**GPIO33**
relay ground	**GND**
MQTT client	**Wi-Fi**

purple
orange
blue
white
GND

Inland 4-ch relay (5V, SRD-05VDC)
loads wired on NC contacts = fail-safe

VCC	5V
IN1	relay_parking_lot
IN2	relay_exterior_signs
IN3	relay_hvac_shed_stage1
IN4	relay_spare_ch4
GND	tied to ESP32 GND

Loads on NC contacts
stay ON if ESP32 or server is down
- Parking lot lights (IN1)
- Exterior signs (IN2)
- Spare (IN4)

IN3 relay_hvac_shed_stage1
(from Andy's notes - kept for now)
Rev 2+: per-RTU stage-1 shed may move to each RTU node's own relay

SAFETY

- **NEVER** put 24VAC on an ESP32 pin (use optocouplers)
- 24VAC only on optocoupler inputs + buck input
- **C required**, taken at the unit (never at thermostat)
- ~1A inline fuse on our transformer feed
- Common GND: relay GND tied to ESP32 GND
- Loads on NC contacts = fail-safe
- Bench only. **No compressor control.**
- Original thermostat stays primary

Wi-Fi MQTT
bkems/{deviceKey}/telemetry (up)
bkems/{deviceKey}/cmd (down)
deviceKey = store-{id}-gw

internet

Raspberry Pi gateway
Mosquitto broker + edge agent
raspberrypi.local / 10.0.0.165

Command Center website
Mac mini now (LAN)
bkems.info later

Wi-Fi MQTT
RTU node)

EACH ROOFTOP UNIT (up to 5)

Original thermostat stays PRIMARY
R W Y1 G C (24VAC)
+ Y2 if the unit has 2-stage cool; W2 if available

Y1 / Y2 cooling-call wires
(each through its own shed relay NC)

RTU's own 24VAC transformer
Tap after any existing R-side fuse/breaker.
C required — taken at the unit (buck return + opto ref).
Never needed at the thermostat.

24VAC R/C

~1A INLINE FUSE
on our feed from the RTU transformer
(protects unit control fuse from our short)

~1A fuse

24VAC to 5VDC buck converter
AT the RTU (pick 5V out, ≥36VAC / ~50VDC in, 1A+)
fed through the ~1A fuse from transformer R/C

W Y1 Y2 G
(24VAC vs C)

24VAC optocouplers (x4)
W, Y1, Y2 (*optional*), G + W2 (*optional*)
AC side referenced to unit C; logic side to RTU-node inputs

Y1 / Y2
in series

SHED RELAYS (5V, NC contacts, *proposed*)

Stage-1 shed: NC in series with Y1

Stage-2 shed (*optional, 2-stage units*): NC in series with Y2

Staging: Y1 first; Y2 only if Y1 can't carry the load.
Y2 is never energized without Y1.
Shed: Y2 first, then Y1. **Restore:** Y1 first, then Y2.
300 s anti-short-cycle applies per stage.

ESP32 RTU NODE - one at each RTU
x up to 5, all pins *proposed*

Wi-Fi	MQTT client
GPIO4	DS18B20 supply/return (optional)
5V / VIN	from RTU buck
3V3	DS18B20 + pull-ups
GND	common GND
GPIO32	W heat call (pull-up)
GPIO33	Y1 cool stage 1 (pull-up)
GPIO27	Y2 cool stage 2 (*optional*, pull-up)
GPIO25	G fan call (pull-up)
GPIO13	W2 heat stage 2 (*optional*)
GPIO26	stage-1 shed relay IN (Y1)
GPIO19	stage-2 shed relay IN (Y2, *optional*)

DS18B20 (optional)
supply / return air
4.7k pull-up to 3V3

5VDC + GND to room plate (CAT5 run if needed)

ROOM SENSOR PLATES x4 (one per RTU)

ESP32-C3 + DS18B20

Green = fan **Red = heat** **Blue = cool**

screw terminals: 5V + GND only

5V fed from the buck AT the RTU (keep ≥4.5V at plate)

Wi-Fi MQTT to the Pi